

Data Analysis (MACSIN4A2225)

Hugo ALATRISTA-SALAS

Final project

1. Introduction

Information systems can be used to model many phenomena. The modeling task begins with the identification of requirements, including information requirements. Then, the objects that simplify reality are identified and described by their attributes or characteristics. These are then stored in data repositories (databases, data warehouses, data lakes, spreadsheets, etc.). Later, instances of the objects are created, and the database is populated. Data is essential to extract information to better understand the phenomena we are studying and to build indicators to support decision-making (Data-Driven Decision Making).

In this context, in this final project, the techniques learned in this course will be used to extract and visualize helpful information from a complex data source.

2. Project goal

The aim of this project is to create a visualization prototype showing key information extracted from the available dataset. To achieve this, we will use the Knowledge Discovery in Databases (KDD) process to analyze and extract useful information from a real multidimensional dataset ¹. Each method or technique used for this purpose should be implemented in a function — using *def name_function():* — and add the parameters you think are necessary. Of course, you can reuse or improve methods implemented in labs during previous sessions. Finally, all the functions created must be called from a *__main__* function implemented at the end of the project, which also launches the visualization prototype. See the code box at the end of this PDF file for inspiration to write your notebook. It is important to note that the material from this course (slides, labs, and Python code) can be used for the final project.

3. About the dataset

As mentioned in the project goal, we will analyze a **multidimensional dataset**. Remember that a multidimensional dataset is composed of three dimensions: 1) temporal, 2) spatial, and 3) analytical. The temporal dimension stores **when** an event occurs; the spatial dimension stores **where** an event occurs; and the analysis dimension records an event's characteristics. The first step of this project is to select a multidimensional dataset according to the team's interests. Various websites provide datasets for learning data science, and some of them offer multidimensional datasets. Some of these websites are:

- Kaggle: <https://www.kaggle.com>
- EM-DAT: <https://doc.emdat.be>
- Google dataset search tool: <https://datasetsearch.research.google.com>

¹<https://www.kdnuggets.com/gspubs/aimag-kdd-overview-1996-Fayyad.pdf>

- Taxi dataset: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- Miscellaneous datasets: <https://github.com/Call-for-Code/logistics-weather-geo/wiki/DataResources>
- NOAA’s archived environmental data: <https://www.ncei.noaa.gov/access/search/index>
- Humanitarian data: <https://data.humdata.org/dataset>

4. Tasks and its evaluation

The evaluation of this project is divided into positive and negative points. The following list describes how the points should be given, considering the main tasks to be implemented in this challenge.

1. [2 points] Create a notebook, upload the data found by the team and store it in a structure that allows data manipulation. Explore the data in a way that allows you to fully understand it (using the data summarization methods provided by Python). This stage should at least show the number of columns and the data type in each column, the number of rows and columns in the dataset, the number of missing data per column and the range of values for each variable. Also include a markdown that **describes the meaning** of the dataset used for this project in **human-readable text**.
2. [7 points] Information building: Create three queries to extract information from the selected dataset. Explain in detail what each query represents and visualise the results. These queries will be distributed as follows:
 - [1 points] A query using grouping queries from the dataset, for example, “the maximal number of ...” or “the sum of the ...”
 - [2 points] A query that uses a frequent pattern mining techniques or other techniques you learn in other courses (*e.g.*, deep learning, text mining, etc.).
 - [4 points] Two queries using temporal or spatial data features (one of each, if the dataset allows). In this section, you could apply the forecasting and spatial clustering techniques covered in the course. You can also apply techniques learnt in other courses, as well as the visualisation.

Note. All indicators should be explained in a *markdown* box: how they are calculated, what they represent, and how they will be interpreted.

3. [2 points] A visualization dashboard: Use Python Dash (<https://dash.plotly.com/>) or another visualization tool to organize the results of the four queries performed in the previous step. The dashboard should include the names of the team members and the name of the dataset used for the project.
4. [1 point] Github repository: Organize and store your code, data and results into a public repository on GitHub.
5. [8 points] Project presentation and demo: The final step is scheduled for the week of June 15. During this time, groups will give 5-minute presentations. All team members will participate. Slides will be used for this purpose. Then, a demo using the notebook uploaded to the DVLC platform will be presented. One student, selected by the lecturer, will present the demo. This means that all members will master the project implementation. The session will conclude with a Q&A.
6. [1 extra point] External dataset: You can enrich your result with an external dataset. You can also use it to interpret the indicators.

If all the previous tasks are completed correctly (including the project presentation and demo), the team will score 20 points (+1 extra point). However, some mistakes can penalize the final score. The most common errors are:

1. [-2 points] Failure to organize all the code into functions and organized into a GitHub repository with a informative *readme* file.
2. [-2 points] If at least one of the functions does not contain comments about its name, input, and output (see example in the code box). Remember also to add comments to make your code more understandable and facilitate the correction of the notebook. If you decide to remove a feature or characteristic of the dataset, you will need to explain why. All decisions should be documented in your proposal.
3. [-2 points] If the dashboard does not work or there is no evidence that it works. Remember to send an HTML or PDF file of the dashboard or other evidence.
4. [-2 points] Each indicator should be described in human terms in a text cell (*markdown*). The penalty for each indicator description is -2 points.
5. [-8 points] If a team member does not explain or answer the questions correctly.

5. Deliverables

1. The notebook in *ipynb* format. The groups must show how they carried out the steps described above (see Section 2) in their notebooks. The grade for this work will depend on how creative they are and how well they follow the instructions given earlier. Comments describing key lines of code are also essential in this submission. This file should be named as *familyNameLeader_notebook.ipynb*.
2. The dashboard in HTML or PDF format: if it is necessary, you can join another file named *familyNameLeader_dashboard.html* or *familyNameLeader_dashboard.pdf*. This file should prove that the dashboard works and clearly show the indicators proposed in this project.
3. A presentation (slides in PDF format) that describe the most important information needed to understand the project. This presentation should be used during the demo and Q&A stage. This file should be named as *familyNameLeader_slides.pdf*.

6. Important information

This work must be completed by groups of at most **six (06) members**. Each team must elect a leader who will upload the project deliverables. It is important to note that only one submission is necessary. This is a team project (there are no individual grades), which means that all group members must be present in the lab throughout the project presentation. At the end of the presentation, the lecturer will ask the group members about their work development. If a group member is unavailable to respond, the group will automatically be penalized by **-8 points**. Remember that this is a group grade, so all members are responsible for actively participating in the development of the work.

Use of LLM: The use of LLMs is permitted in this project. If any part of the code has been developed using an LLM, please provide the prompt in a *markdown* box.

familyNameResponsable_notebook.ipynb

```
# Team members and GitHub link
FirstName, FamilyName (team head)
FirstName, FamilyName (team member)
Link to GitHub repository
...

# Import libraries
import pandas as pd
import numpy as ny
...

# Function definitions

# Function for reading a CSV file
# Input: the path to the csv file
# Output: a dataframe
def load_data(file_path):
    bla bla bla # Try to comment on each line
    bla bla bla # especially if this line helps us to understand your project
    ...

# Function to bla bla bla
# Input: bla bla bla
# Output: bla bla bla
def function_1(df):
    bla bla bla
    ...

def function_2(df):
    bla bla bla
    ...

# Main block
if __name__ == "__main__":
    # Step 1: Data collection
    file_path = "myfile.csv"
    data = load_data(file_path)
    ...

    #Step 2: Indicators construction
    result = function_1(df) # this is only an example
    ...
```

May the force be with you !!!